PROJECT AND TEAM INFORMATION

Project Title

AIRPORT RUNWAY AND FLIGHT MANAGEMENT SYSTEM

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Although there are very few runways in any airport, the planes cannot land or take off one after another,and several of them might request to use the same runway simultaneously. processing such requests on the basis of their order of arrival would mean that an aircraft with an urgent situation would have to wait for a regular plane to land,which is unacceptable. We aimed to create a system whose logic of operation should reflect the process in question: priority first and then the arrival time. And even the basic simulation of this problem would be clearly to illustrate the importance of priority based scheduling

State of the Art / Current solution (1 pt)

This is practically done by air traffic controllers with the aid of information from radar, flights' schedule, and the use of separation procedures, aided by automation technologies such as Traffic Collision Avoidance System and arrival and departure sequence automation programs for air navigation authorities. The actual systems take into account factors such as weather conditions, type of aircraft, and its fuel level among others; these are way more complicated than anything we could model on our project. In other areas outside aviation, the closest example of the application of such an algorithm we have all been exposed to is that of CPU scheduling in operating systems, where a priority queue selects the process to be executed next.

Project Goals and Milestones (2 pts)

The main goal is:-
1.To accept flight details for arrival and departure
2.Convert every flight condition(emergency,patient on board,low fuel, vip ,normal into numeric priority 3.The highest priority waiting flight will be served first
4.Assign up to two runways at once while other flights are being placed in a waiting list
5.Automatically allocate the next flight to the vacant runway.
Milestone 1:Define the flight and finalize the priority routes
milestone 2:Implement the priority queue and comparison logic and test it with sample flights
Milestone 3:Add runway allocation, waiting list and automatic reassignment
Milestone 4:Build the menu driven,test with different inputs and prepare the final report

Project Approach (3 pts)

The project will be developed using c++ with data structure and object oriented concepts . Each flight has a struct with the information about its number, type (arrival or departure), condition, and priority. Each file is stored as a struct which will have it flight number ,arrival and departure, condition and priority . The assignRunways() function counts how many of the two runways are available and dequeues flights accordingly. The completeFlight() function makes a runway available and then tries to find the next flight to assign. We opted for such an architecture because it separates runway assignment algorithm from the queue structure operations, which allowed us to build the project piece by piece testing each part.

System Architecture (High Level Diagram)(2 pts)

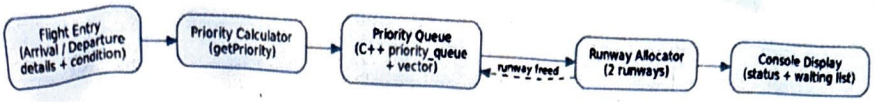


Figure 1:

Project Outcome / Deliverables (1 pts)

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The primary product from this project will be a console-based program that will manage the runway and will schedule arrivals and departures based on priority-
This project will result in:
1. *A compiled console application showing arrivals and departures being scheduled.*
2. *The logic of priority-based runway scheduling with re-scheduling when runway becomes free.*
3. *A data structure used as part of the queue.*
4. *Test cases for the project showing its performance in different scenarios.*
5. *A written report describing the problem and priorities.*

Assumptions

There are assumed to be only two runways available in the entire simulation, that the status of the flight is known at the time when it is entered into the queue, and that the priority of a flight does not change after it is entered into the queue. Weather and aircraft separation issues are out of scope of the project.

References

C++ Reference - https://en.cppreference.com/w/cpp/container/priority$_q$ueue
GeeksforGeeks — https://www.geeksforgeeks.org/priority-queue-set-1-introduction/
Federal Aviation Administration — https://www.faa.gov/air$_t$raffic